

Email Sending Project using python (single user)

Project Goal: Email sending for a Single User

You will create a Python script that can send an email to someone (like yourself or a friend) with a custom subject and message using your Gmail account

Demo Example

Suppose you want to send an email like this:

- **To:** yourfriend@example.com
- **Subject:** Meeting Reminder
- **Body:**

Hi Guys ,

Just reminding you about our meeting tomorrow at 10 AM.

Regards,

Jani

This script will send it, instead of you opening Gmail manually every time.

Tools Needed

- Python installed on your system.
- A Gmail account.
- Enable Gmail's "App Password" (safer than using your real password).

What is smtplib?

Full Form:

smtplib = SMTP Library

- It stands for: **Simple Mail Transfer Protocol Library**
 - It is a **built-in Python module** used to send emails using the **SMTP protocol** (the standard protocol for email transmission).
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What is smtplib.SMTP_SSL(...)?

◆ Full Form:

SMTP_SSL = SMTP over Secure Sockets Layer

It is a **secure version of the SMTP client**, which:

- Uses **SSL (Secure Sockets Layer)** to encrypt the connection between your Python script and the email server (like Gmail).
 - Ensures your **login and email content are safely transmitted**, protecting from hackers.
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Summary Table

Term	Full Form	Purpose
smtplib	Simple Mail Transfer Protocol Library	Python module to send emails
SMTP_SSL()	SMTP over Secure Sockets Layer	Creates a secure connection to email server

Step-by-Step Project Development

Step 1: Setup Gmail for Sending Emails

1. **Enable 2-Step Verification** on your Gmail:

- Go to: <https://myaccount.google.com/security>
- Turn on **2-Step Verification**

2. **Generate App Password:**

- After enabling 2FA, go to:
<https://myaccount.google.com/apppasswords>
- Select **Mail** and **Windows Computer**, click **Generate**.
- Copy the 16-character password shown (you'll use it in the Python script instead of your Gmail password).

Step 2: Install Python and Create Script

No need to install anything extra. Python comes with the required libraries.

Let's write the code now:

Code:

```
import smtplib
```

```
from email.message import EmailMessage
```

```
# Email details
```

```
sender_email = 'janpadushaik@gmail.com'
```

```
receiver_email = 'janibashadshaik@gmail.com'
```

```
app_password = 'ktsi quvu syfy jyjh '
```

```
# Create email content
```

```
subject = 'Meeting Reminder'
```

```
body = '''Hi Guys
```

```
Just reminding you about our meeting tomorrow at 10 AM.
```

```
Regards,
```

```
Jani'''
```

```
# Create email message
```

```
msg = EmailMessage()
```

```
msg['Subject'] = subject
```

```
msg['From'] = sender_email
```

```
msg['To'] = receiver_email
```

```
msg.set_content(body)
```

```
# Send the email
```

```
try:
```

```
    with smtplib.SMTP_SSL('smtp.gmail.com', 465) as smtp:
```

```
        smtp.login(sender_email, app_password)
```

```
        smtp.send_message(msg)
```

```
        print("Email sent successfully!!")
```

```
except Exception as e:
```

```
    print("Error:", e)
```

Sample Output

Email sent successfully!

Or if there's an error (e.g., wrong password):

Error: SMTPAuthenticationError ...

Explanation (Line-by-Line)

Line	What it does
<code>import smtplib</code>	Handles email sending protocol
<code>from email.message import EmailMessage</code>	Helps create the email format
<code>msg['From'], ['To'], ['Subject']</code>	Sets email fields
<code>msg.set_content(body)</code>	Adds your custom message
<code>with smtplib.SMTP_SSL(...)</code>	Connects to Gmail securely
<code>smtp.login(...)</code>	Logs in with your email and app password
<code>smtp.send_message(msg)</code>	Sends the email

Step 3: Run the Script

1. Save your script as `send_email.py`
2. Open terminal or command prompt.
3. Run:

```
python send_email.py
```

Step 4: Customize It (Optional)

You can now:

- Accept user input (input() function) for subject and body.
- Read email addresses from a file.
- Schedule it using schedule or cron.

```
import smtplib  
  
from email.message import EmailMessage  
  
import schedule  
  
import time
```

```
# Email account details
```

```
sender_email = 'yourgmail@gmail.com'
```

```
receiver_email = 'yourfriend@example.com'
```

```
app_password = 'your_app_password_here'
```

```
# Function to send email
```

```
def send_email():
```

```
    subject = 'Daily Reminder!'
```

```
    body = '''Hi friend,
```

```
Just sending you a daily reminder as planned.
```

```
Have a great day!
```

```
– Jani'''
```

```
# Create email
```

```
msg = EmailMessage()
```

```
msg['Subject'] = subject
```

```
msg['From'] = sender_email
```

```
msg['To'] = receiver_email
```

```
msg.set_content(body)
```

```
# Send email
```

```
try:
```

```
with smtplib.SMTP_SSL('smtp.gmail.com', 465) as smtp:
    smtp.login(sender_email, app_password)
    smtp.send_message(msg)
    print("Email sent successfully at 10 AM!")
except Exception as e:
    print("Error sending email:", e)

# Schedule the email every day at 10:00 AM
schedule.every().day.at("10:00").do(send_email)

print("Scheduler started... Will send email every day at 10 AM.")

# Keep running the scheduler
while True:
    schedule.run_pending()
    time.sleep(60) # check every 60 seconds
```